

BORDER OPTICS

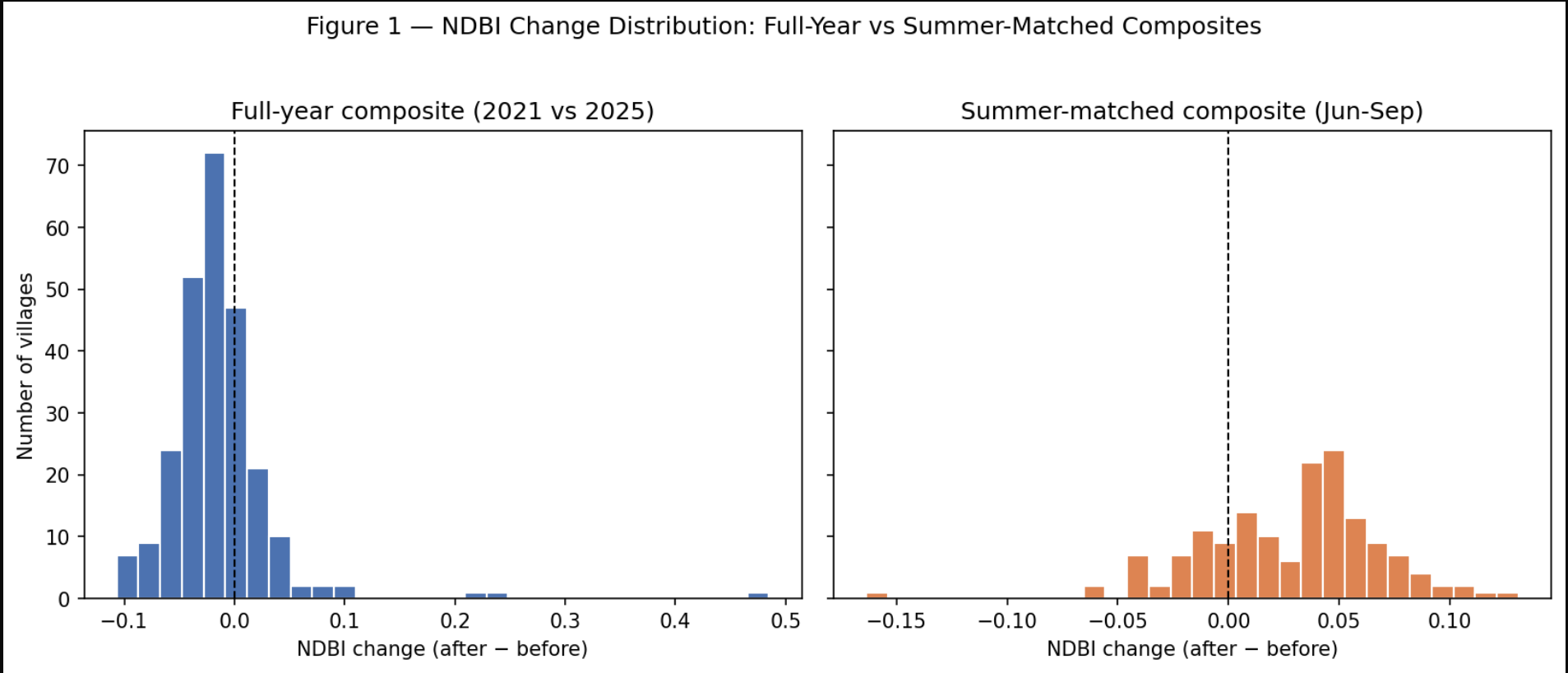
Maps & Plots

Satellite-Based Verification Framework for India's Vibrant Villages Programme (VVP-I)

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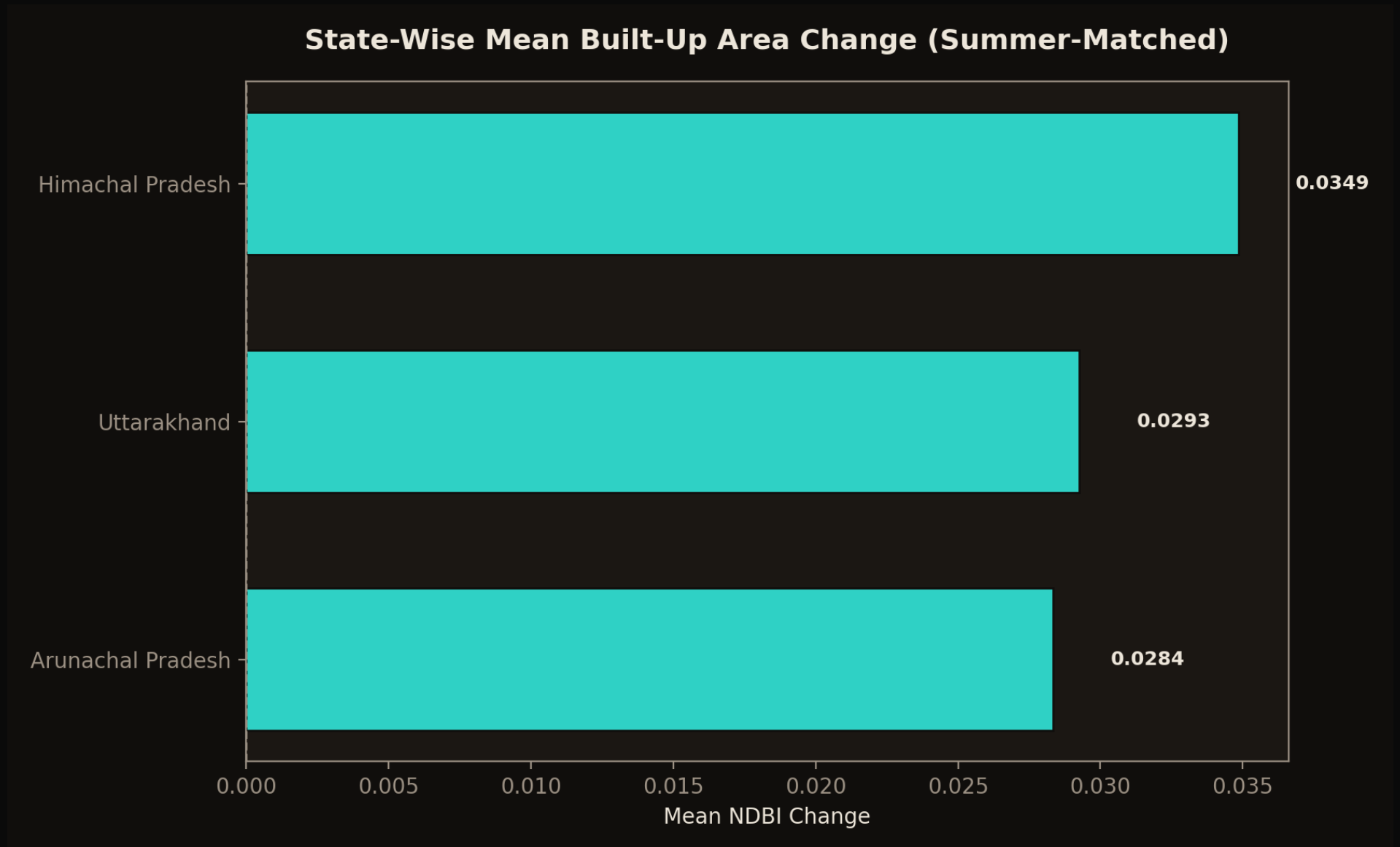
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Figure 1 — NDBI Change Distribution



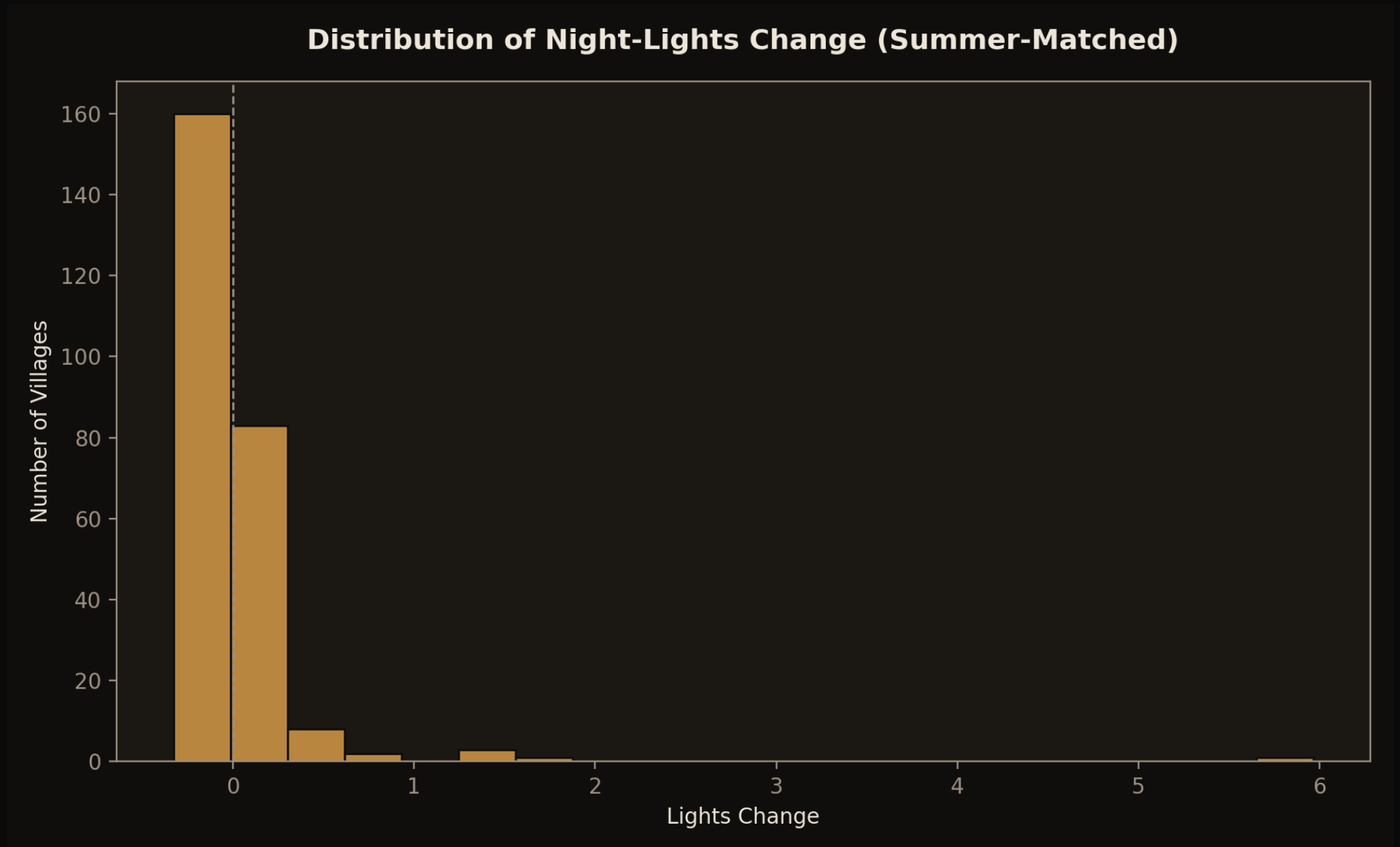
Full-year vs summer-matched composites, all geocoded villages.

Figure 2 — State-Wise Mean Built-Up Area Change



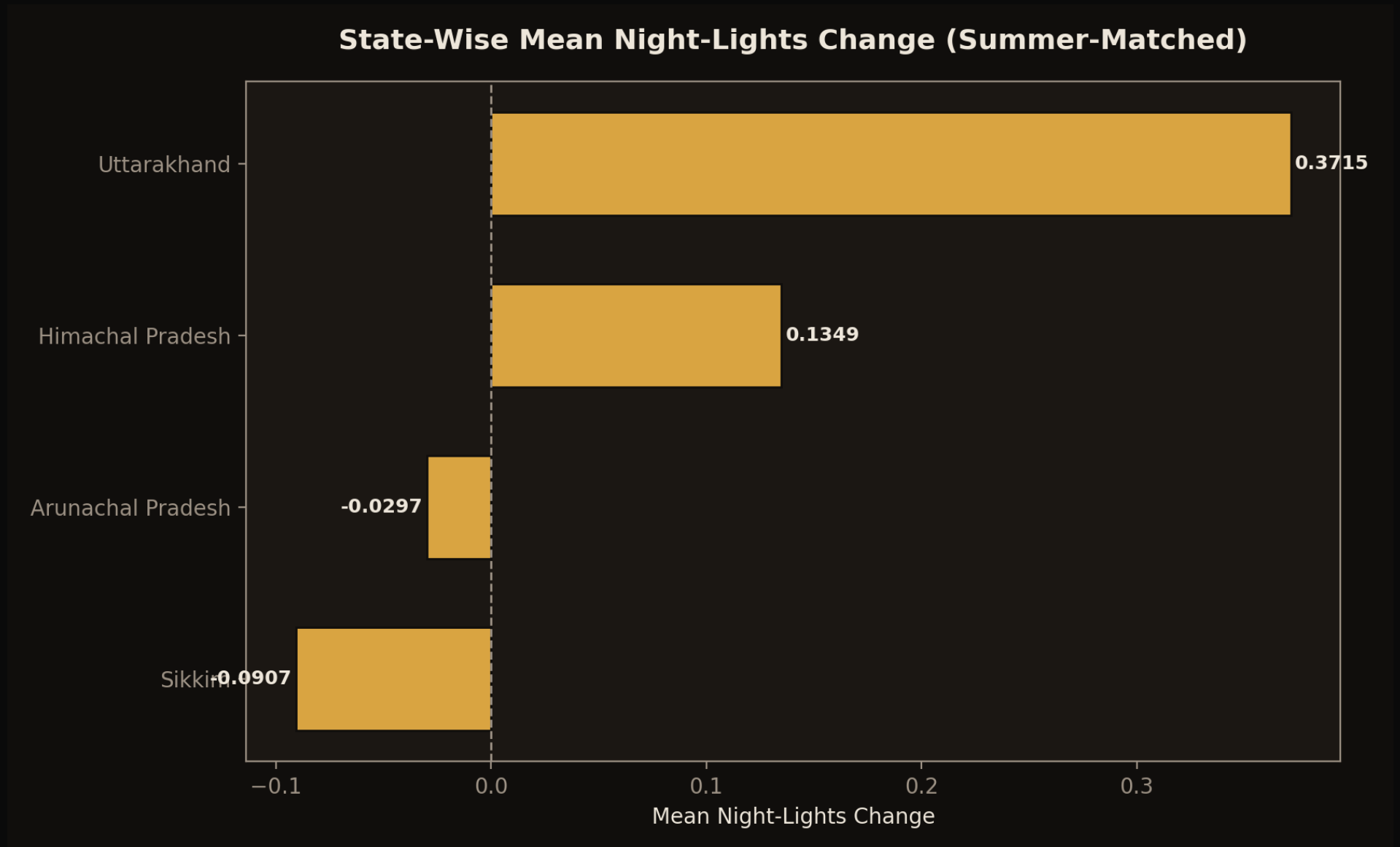
Summer-matched composite, by state.

Figure 3 — Distribution of Night-Lights Change



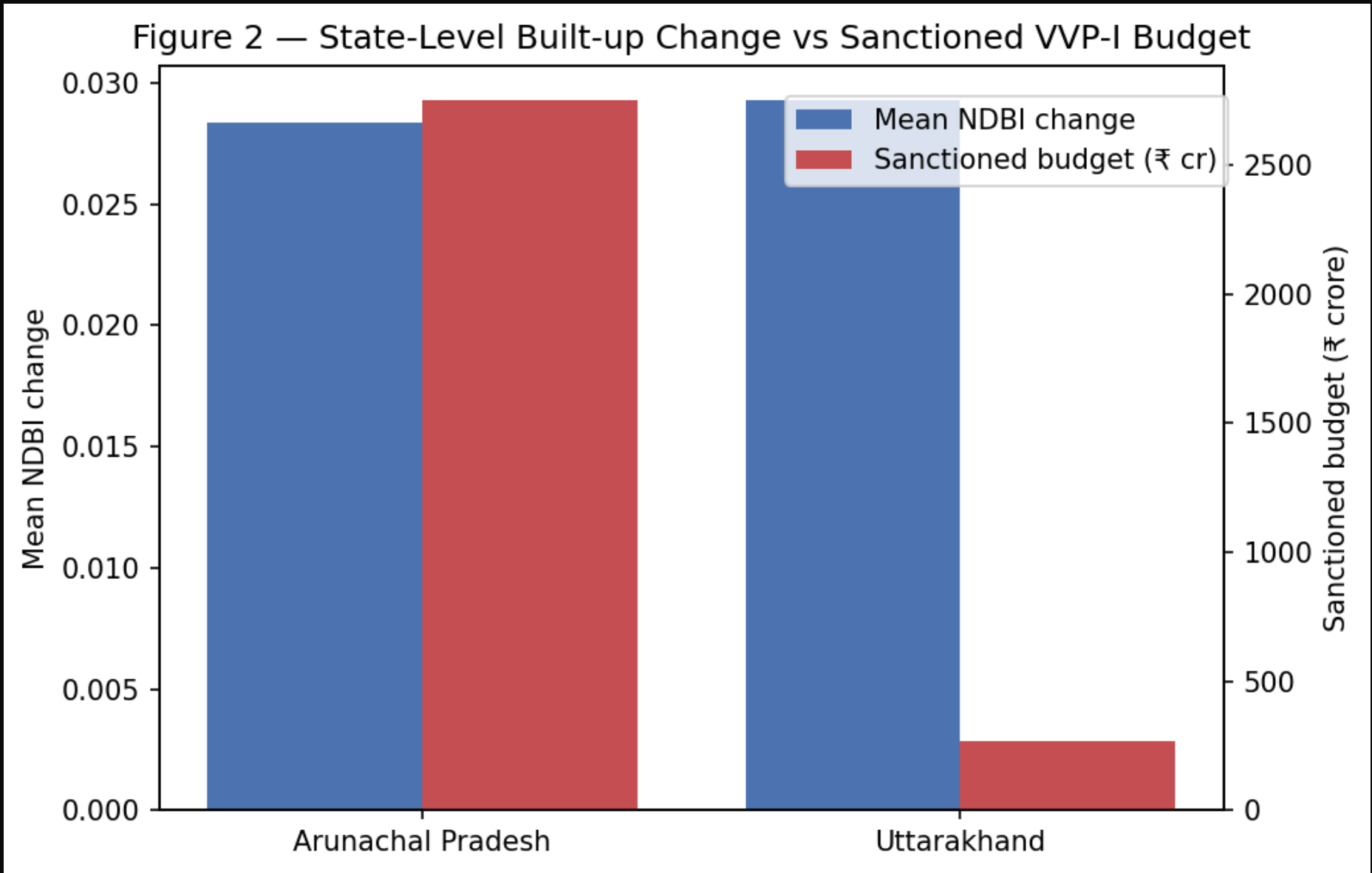
Summer-matched composite, all geocoded villages.

Figure 4 — State-Wise Mean Night-Lights Change



Summer-matched composite, by state.

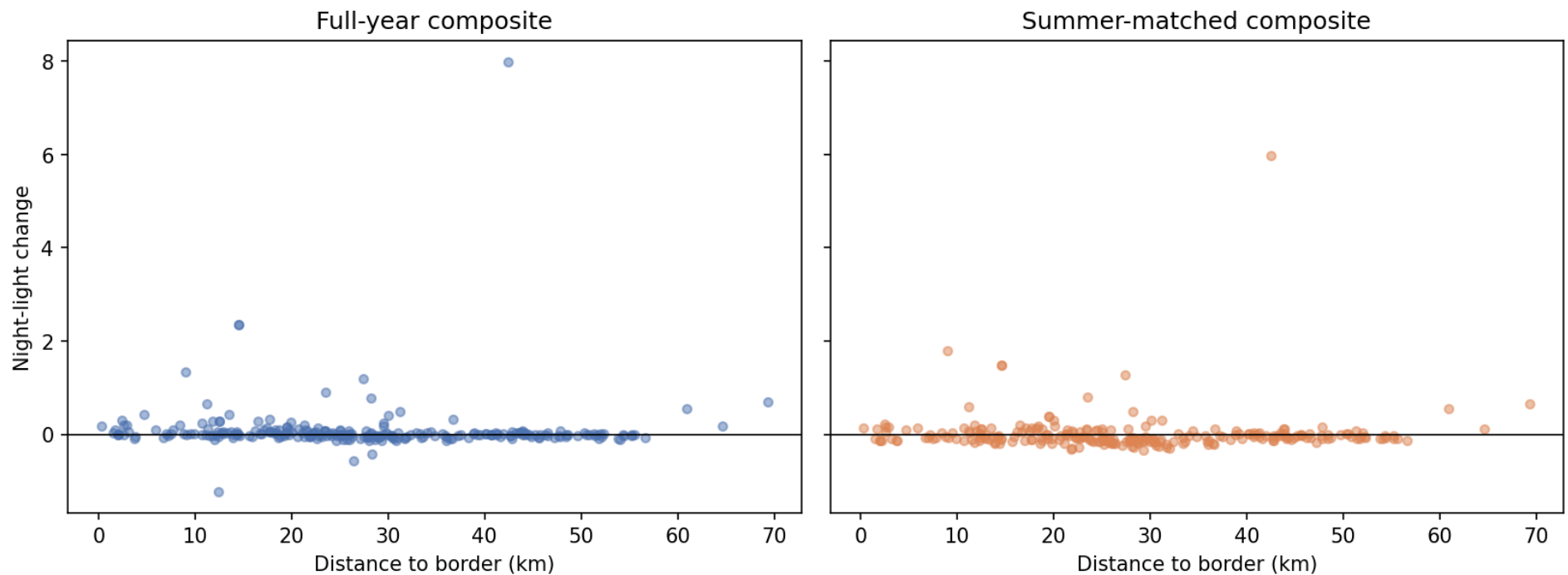
Figure 5 — State-Level Built-up Change vs Sanctioned Budget



Summer-matched NDBI change, Arunachal Pradesh vs Uttarakhand (RQ2).

Figure 6 — H3: Border Distance vs NDBI and Night-Light Change

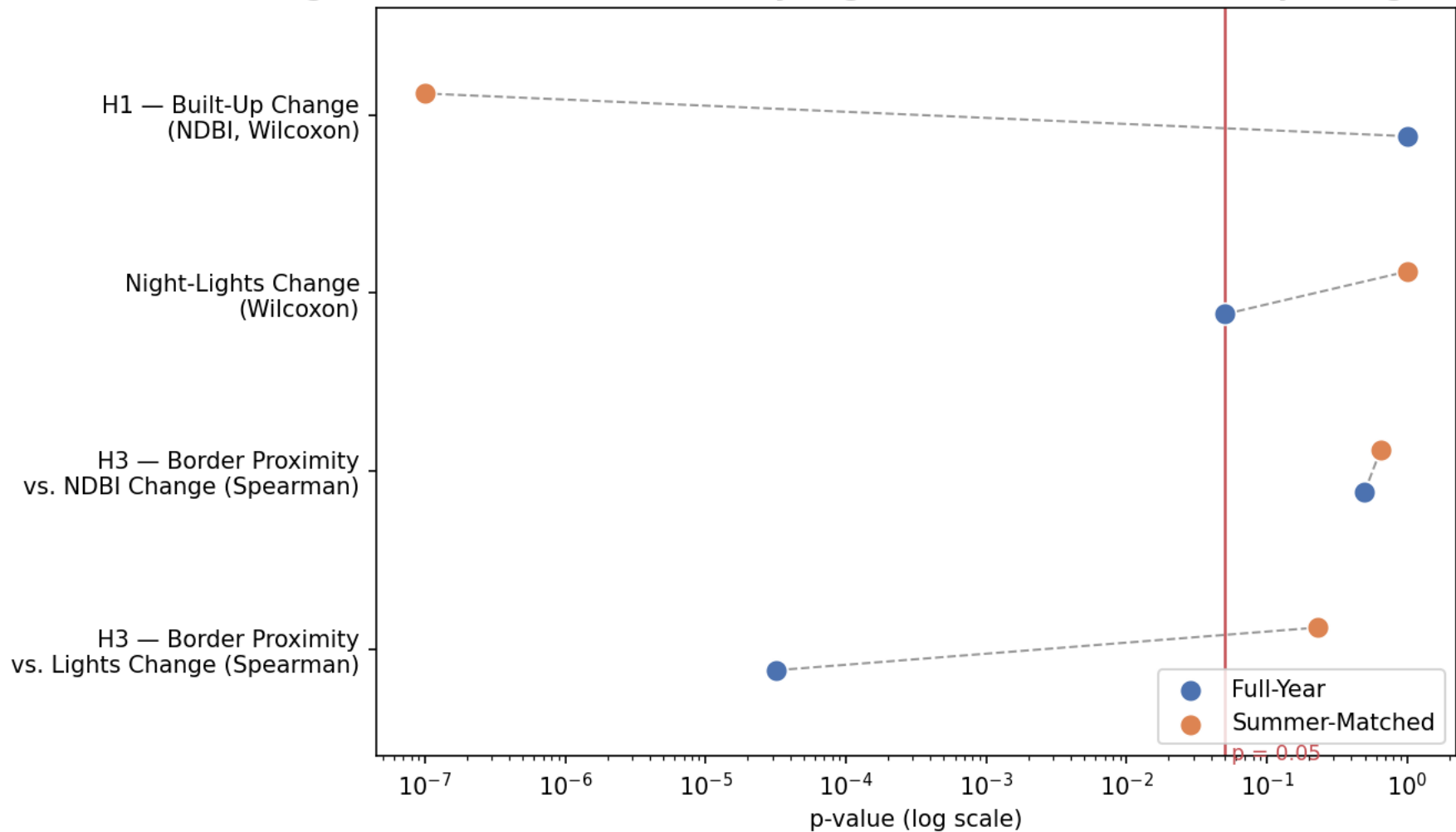
Figure 3 — H3: Border Distance vs Night-Light Change (Full-Year vs Summer-Matched)



Full-year vs summer-matched composites.

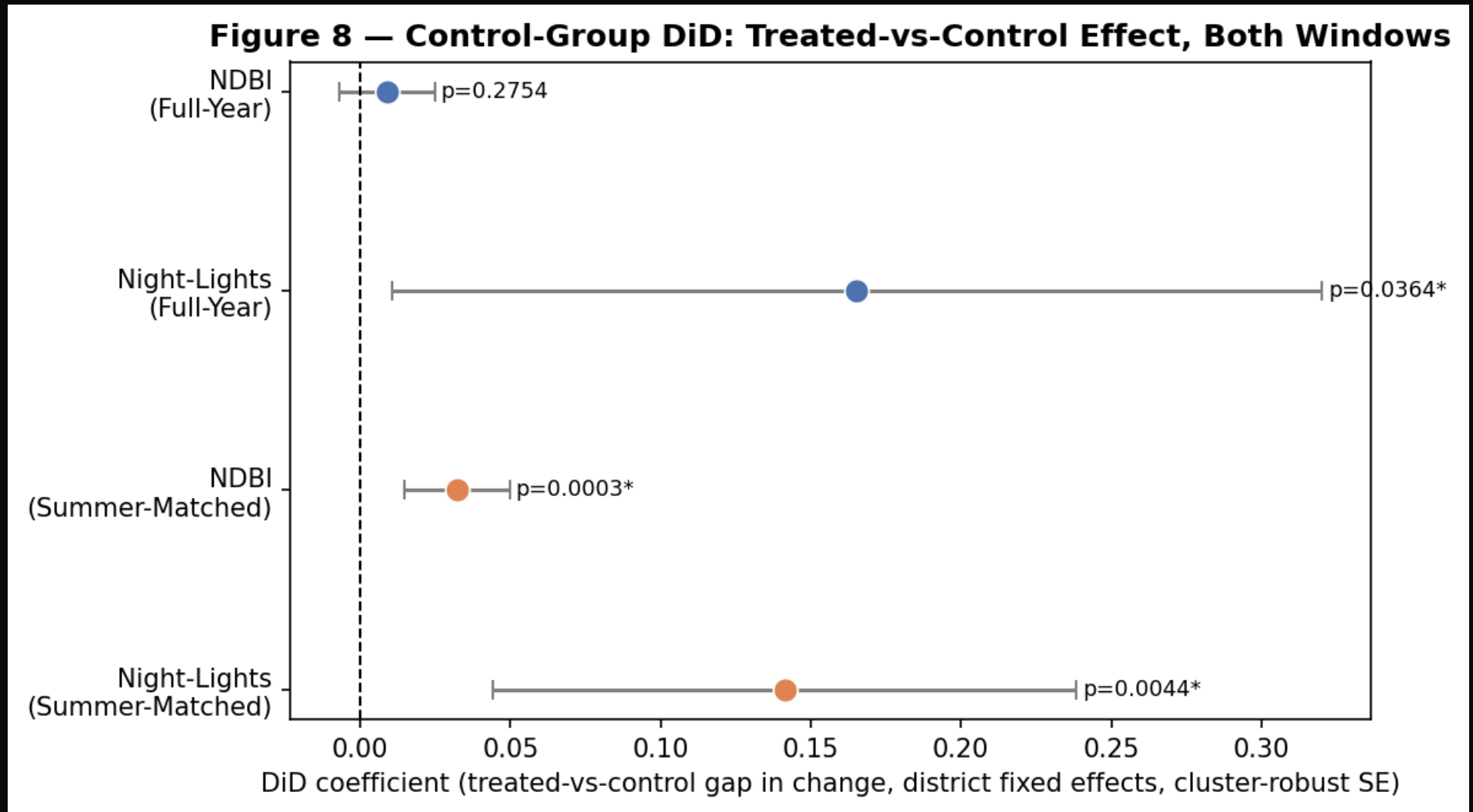
Figure 7 — Robustness Summary

Figure 7 — Robustness Summary: Significance Across Both Compositing Windows



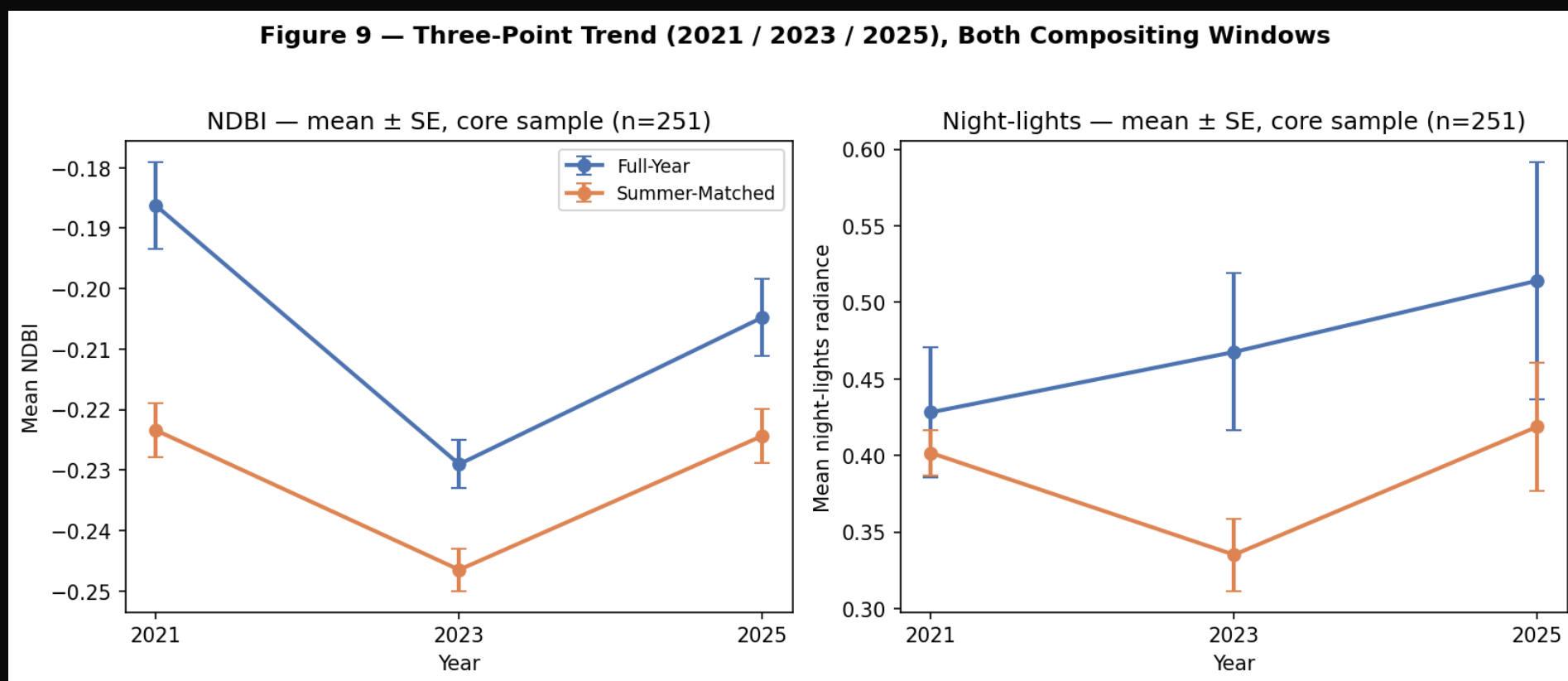
Significance of all four core tests under both compositing windows.

Figure 8 — Control-Group Difference-in-Differences Effect



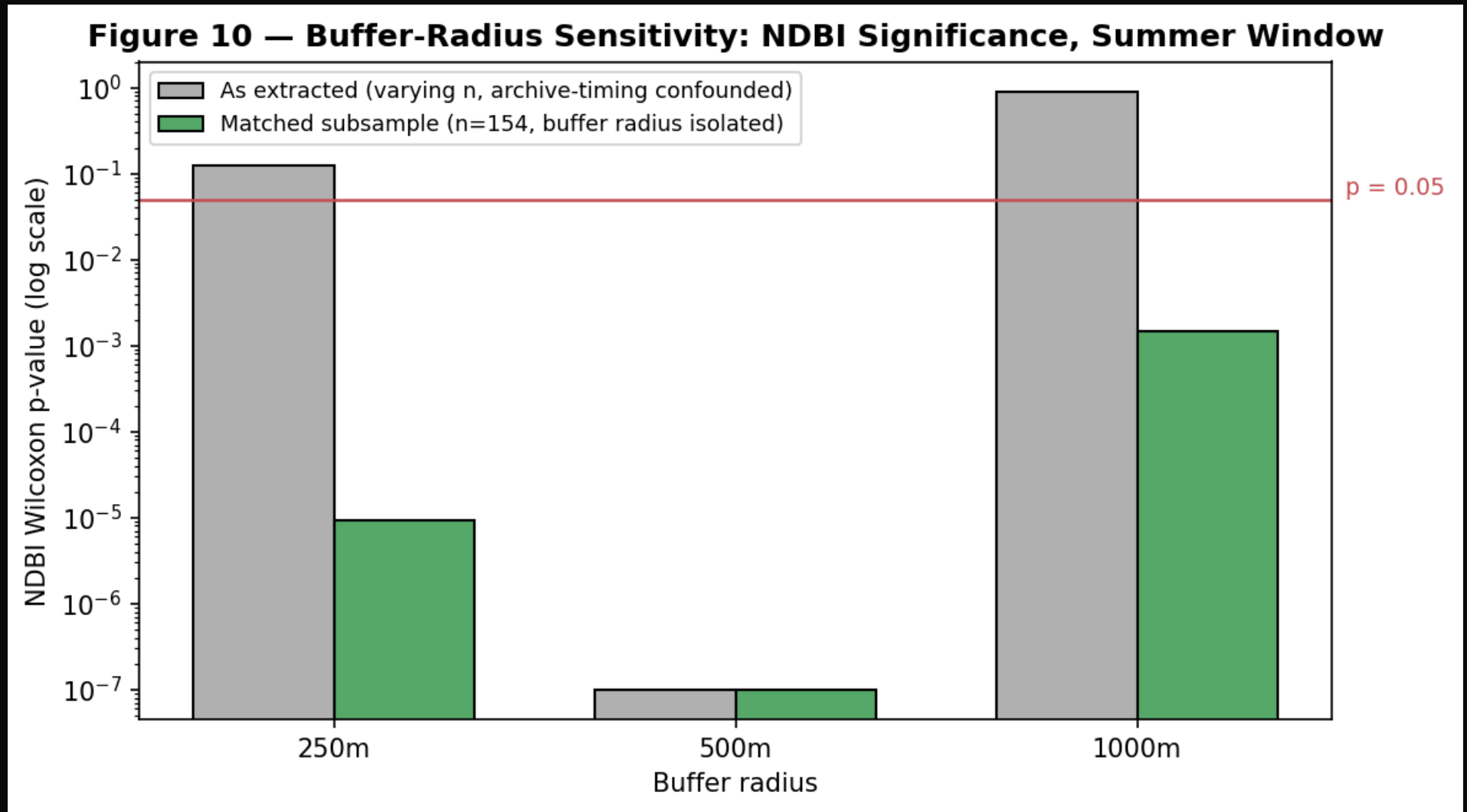
Treated-vs-control gap in change, district fixed effects, both compositing windows (H4).

Figure 9 — Three-Point Multi-Year Trend



Mean NDBI and night-lights radiance at 2021, 2023, and 2025, both compositing windows.

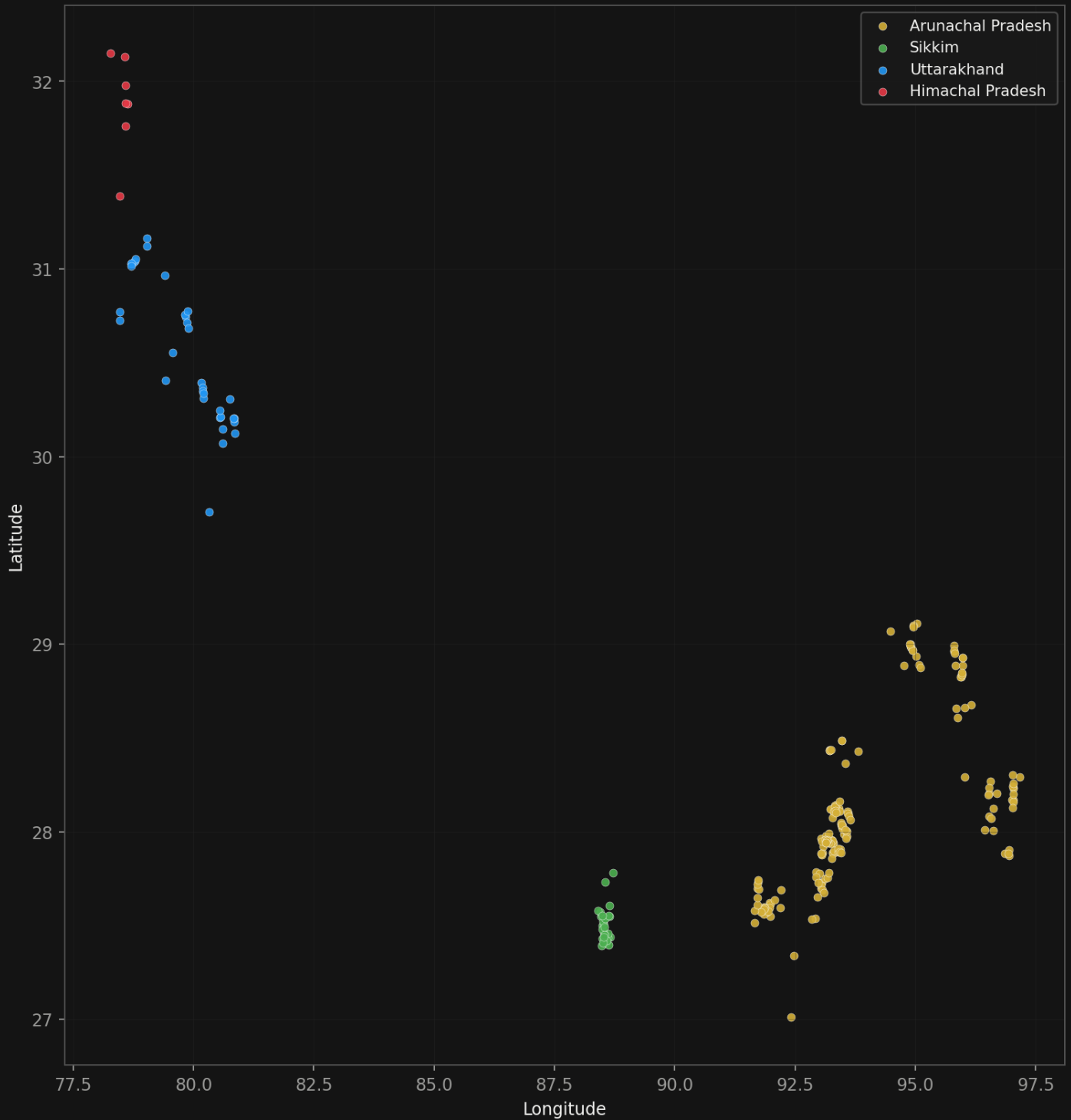
Figure 10 — Buffer-Radius Sensitivity



NDBI significance at 250m, 500m, and 1km buffer radii, summer window.

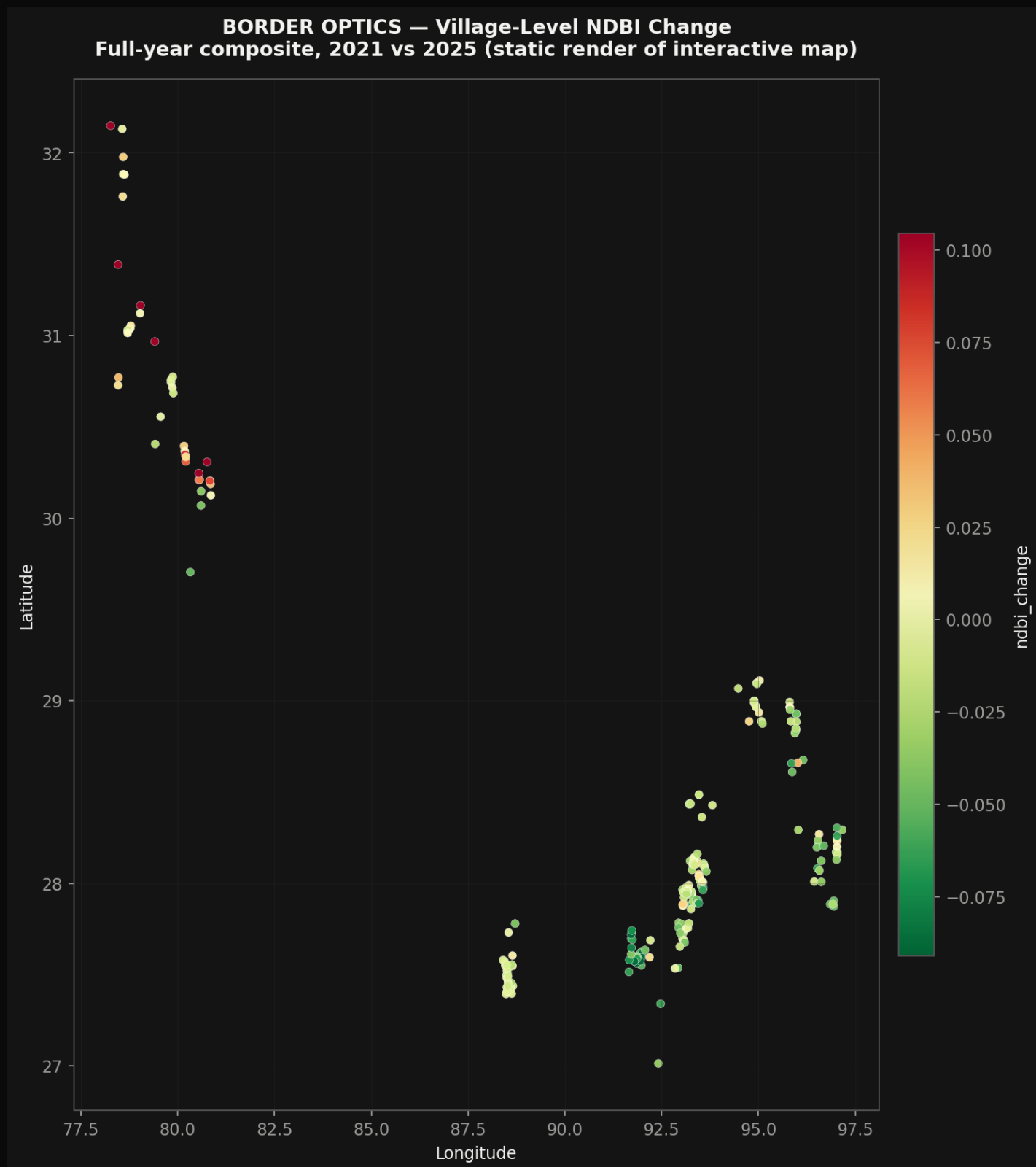
Map 1 — Village Sample Overview

BORDER OPTICS — Village Sample Overview
All geocoded villages across 4 states, colored by state (static render of interactive map)



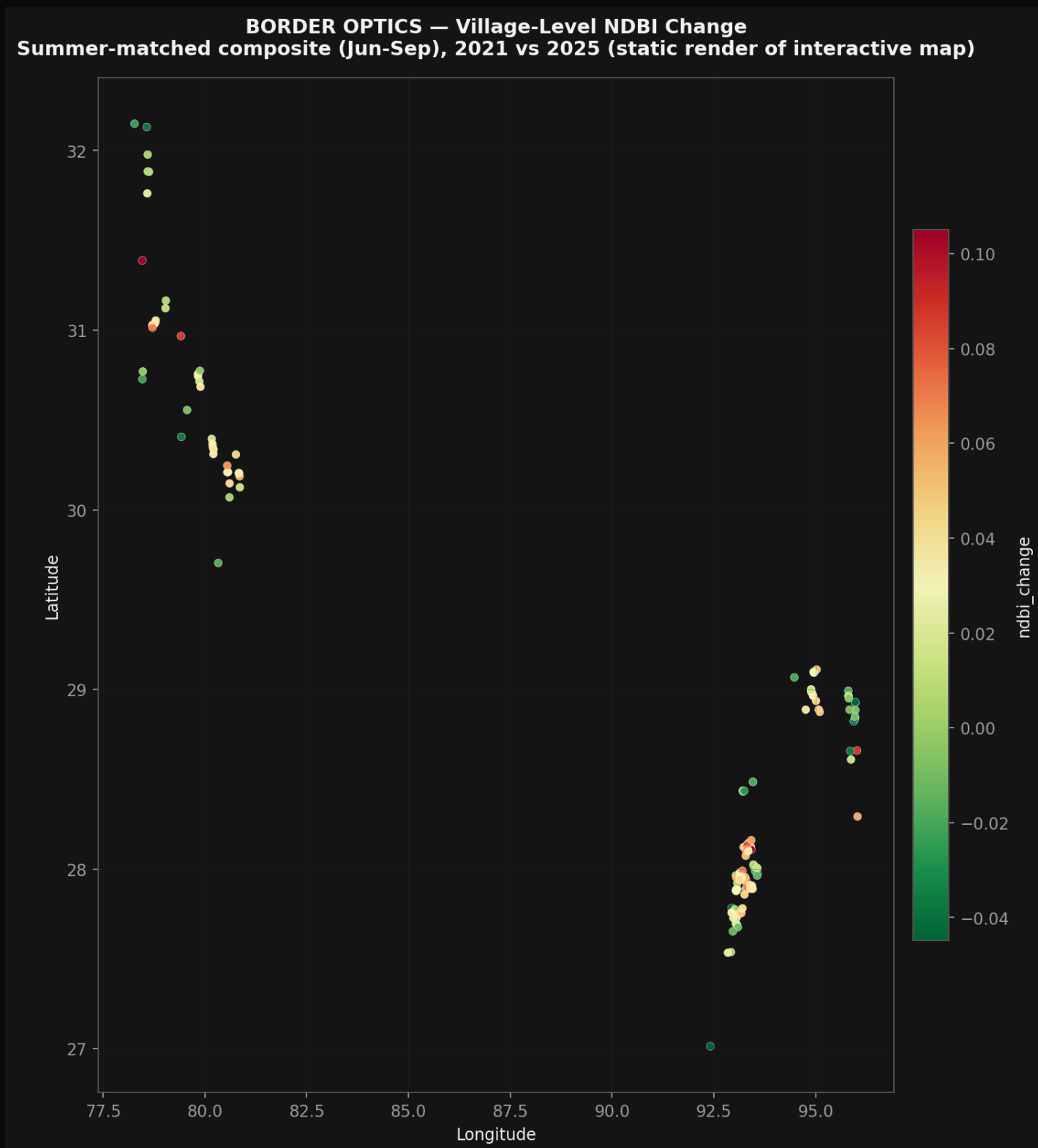
All 258 geocoded villages, colored by state. Interactive version: [dashboard](#) → [Interactive Maps](#) → [State Overview](#).

Map 2 — NDBI Change (Full-Year Composite)



Interactive version: [dashboard](#) → [Interactive Maps](#) → [Built-Up Change](#) — Full-Year Composite.

Map 3 — NDBI Change (Summer-Matched Composite)



Interactive version: [dashboard](#) → [Interactive Maps](#) → [Built-Up Change](#) — Summer-Matched Composite.

Map 4 — Night-Lights Change



Interactive version: [dashboard](#) → [Interactive Maps](#) → [Night-Lights Change](#).

Map 5 — Village Proximity to Border/LAC

BORDER OPTICS — Village Proximity to Border/LAC
Straight-line distance to nearest Natural Earth boundary segment (static render of interactive map)

